

KY-002 Vibration Sensor - ESP32 Integration Guide

Document: Technical Documentation

Sensor Model: KY-002 Vibration Switch Sensor Module

Platform: ESP32 Microcontroller

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1. Introduction

The KY-002 is a digital vibration sensor module based on a mechanical vibration switch. It detects shock and vibration events and outputs a digital signal. This sensor is widely used in anti-theft systems, motion detection applications, and environmental monitoring projects.

The module features a simple interface with three pins, making it easy to integrate with microcontrollers like the ESP32. When vibration is detected, the sensor produces a momentary LOW signal that can be captured by the microcontroller's digital input pin.

2. Operating Principle

The KY-002 sensor operates using a mechanical vibration switch that contains a spring contact mechanism. Under normal conditions (no vibration), the internal contacts remain in a stable position, and the output signal is HIGH. When vibration or shock occurs, the spring contacts momentarily close, causing the output to go LOW.

Key Characteristics:

- Digital output (HIGH/LOW logic)

- Momentary contact closure on vibration
- High sensitivity to mechanical shock
- No calibration required

The sensor responds to various types of vibration including impacts, knocks, and continuous oscillations. The response time is very fast, typically less than 1ms, making it suitable for real-time monitoring applications.

3. Technical Specifications

Parameter	Value
Operating Voltage	3.3V - 5V DC
Output Type	Digital (TTL compatible)
Output Signal	HIGH (no vibration), LOW (vibration detected)
Response Time	< 1ms
Operating Temperature	-10°C to +50°C
Module Dimensions	18.5mm × 15mm
Mounting Holes	M3 compatible
Interface	3-pin header (VCC, GND, Signal)

4. Pin Configuration

KY-002 Module Pinout

- 1

GND

Ground (0V) - Connect to ESP32 GND
- 2

VCC

Power Supply - Connect to ESP32 3.3V
- 3

Signal

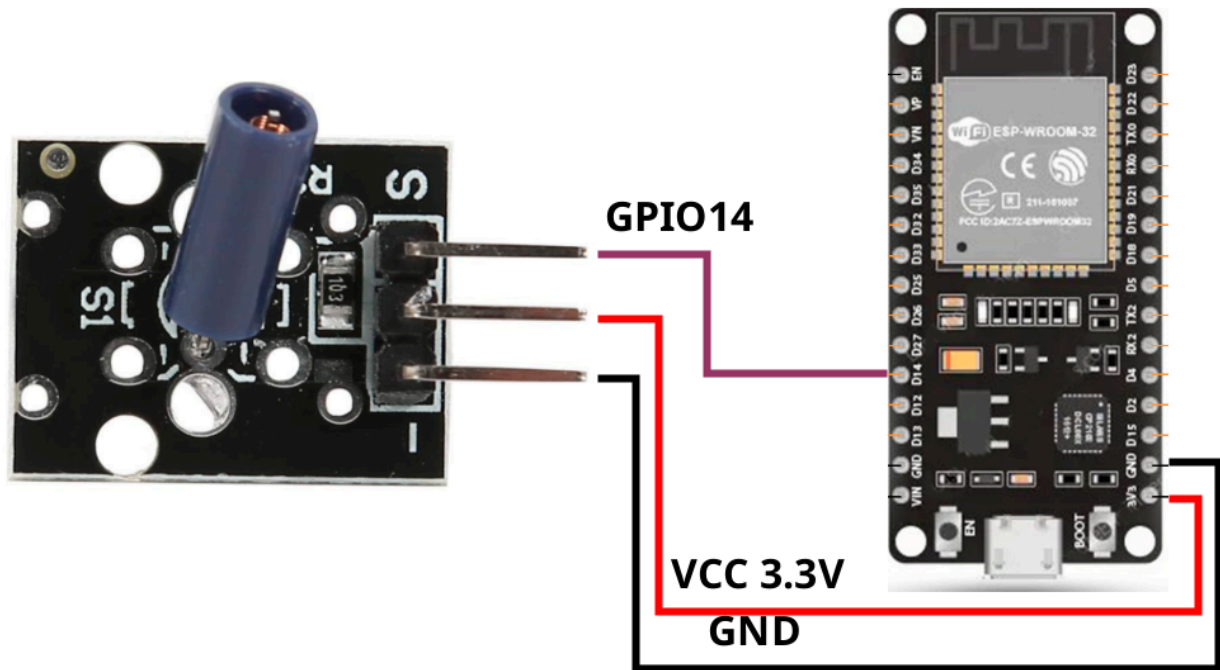
Digital Output - Connect to ESP32 GPIO pin

5. ESP32 Connection

5.1 Wiring Diagram

The KY-002 sensor connects directly to the ESP32 without additional components. The recommended connection is as follows:

KY-002 Pin	ESP32 Pin	Description
GND	GND	Common ground connection
VCC	3.3V	Power supply (3.3V recommended)
Signal	GPIO 14	Digital input with internal pull-up



Note: While the KY-002 can operate at 5V, using 3.3V is recommended for ESP32 to ensure GPIO pin safety. The ESP32 GPIO pins are not 5V tolerant.

5.2 Hardware Setup

1. Connect the KY-002 GND pin to ESP32 GND
2. Connect the KY-002 VCC pin to ESP32 3.3V output
3. Connect the KY-002 Signal pin to ESP32 GPIO 14 (or any available GPIO)
4. Ensure all connections are secure
5. Mount the sensor in the desired monitoring location

6. Programming the ESP32

6.1 Basic Reading Example

This example demonstrates how to read the vibration sensor and display events on the Serial Monitor:

```
// KY-002 Vibration Sensor - Basic Example
#include <Arduino.h>
#define VIBRATION_PIN 14 // GPIO 14
#define LED_PIN 2 // Built-in LED
void
```

```
setup() { Serial.begin(115200); pinMode(VIBRATION_PIN, INPUT_PULLUP);  
pinMode(LED_PIN, OUTPUT); Serial.println("KY-002 Vibration Sensor  
Ready"); } void loop() { int sensorValue = digitalRead(VIBRATION_PIN);  
if (sensorValue == LOW) { digitalWrite(LED_PIN, HIGH);  
Serial.println("⚠ Vibration Detected!"); delay(100); } else {  
digitalWrite(LED_PIN, LOW); } delay(10); }
```

6.2 Event Counter with Debouncing

This advanced example counts vibration events and implements debouncing to avoid false triggers:

```
// KY-002 with Event Counting and Debouncing #include <Arduino.h>  
#define VIBRATION_PIN 14 #define DEBOUNCE_TIME 50 // milliseconds  
unsigned long vibrationCount = 0; unsigned long lastTriggerTime = 0;  
bool lastState = HIGH; void setup() { Serial.begin(115200);  
pinMode(VIBRATION_PIN, INPUT_PULLUP); Serial.println("KY-002 Event  
Counter Ready"); } void loop() { int currentState =  
digitalRead(VIBRATION_PIN); unsigned long currentTime = millis(); if  
(currentState == LOW && lastState == HIGH) { if (currentTime -  
lastTriggerTime > DEBOUNCE_TIME) { vibrationCount++; lastTriggerTime =  
currentTime; Serial.print("Event #"); Serial.print(vibrationCount);  
Serial.println(" - Vibration detected!"); } } lastState = currentState;  
delay(10); }
```

7. Typical Applications

7.1 Anti-Theft Security System

The KY-002 can be mounted on doors, windows, or valuable objects to detect unauthorized access or tampering. When vibration is detected, the system can trigger an alarm or send a notification.

7.2 Earthquake Detection

Multiple KY-002 sensors can be deployed to create a basic seismic monitoring network. While not as precise as professional seismometers, they can provide early warning signals for strong vibrations.

7.3 Machine Vibration Monitoring

Industrial equipment monitoring to detect abnormal vibrations that may indicate mechanical failure or maintenance requirements. The sensor can be attached to motors, pumps, or rotating machinery.

7.4 Knock Detection System

Interactive installations where knocking or tapping triggers specific actions, such as turning on lights, playing sounds, or activating displays.

7.5 Vehicle Security

Detecting impacts or movements of parked vehicles to trigger anti-theft alarms or send alerts to the vehicle owner.

7.6 Package Monitoring

Monitoring shipment packages during transit to detect rough handling or excessive shaking that could damage sensitive contents.

8. Best Practices

Recommendations:

- **Power Supply:** Use stable 3.3V power supply to avoid false triggers
- **Pull-up Resistor:** Enable internal pull-up on GPIO pin for reliable readings
- **Debouncing:** Implement software debouncing to filter noise
- **Mounting:** Secure the sensor firmly to the monitoring surface
- **Sampling Rate:** Use 10ms delay (100 samples/second) for responsive detection
- **Environmental Protection:** Shield sensor from moisture and extreme temperatures

9. Troubleshooting

False Triggers

Problem: Sensor triggers without actual vibration.

Solution: Check power supply stability, add debouncing delay, ensure proper grounding, and verify connections are secure.

No Detection

Problem: Sensor does not respond to vibration.

Solution: Verify wiring connections, check that INPUT_PULLUP is enabled, test sensor with known vibration source, and confirm GPIO pin is not damaged.

Intermittent Response

Problem: Sensor works inconsistently.

Solution: Check for loose connections, verify power supply voltage, clean sensor contacts, and ensure proper mounting.

10. Conclusion

The KY-002 vibration sensor provides a simple and effective solution for detecting mechanical vibrations and shocks. Its digital output interface makes it easy to integrate with the ESP32 platform, enabling a wide range of monitoring and security applications.

By following the connection guidelines and programming examples in this document, developers can quickly implement vibration detection systems with minimal hardware requirements and straightforward software implementation.

ESP32 + KY-002 Vibration Sensor Technical Documentation

For more information and updates, visit the project repository